

SkillBridge — Skill Gap Analyser & Placement Intelligence Platform

A full-stack, AI-powered MERN application that bridges the gap between student skills, institutional placement management, and corporate hiring intelligence through predictive analytics, shared-state Kanban workflows, and curated learning paths.

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Project Overview

SkillBridge is a multi-role SaaS placement intelligence platform designed for the modern university ecosystem. It operates on three distinct personas — **Student**, **College (TPO)**, and **Recruiter** — each with a dedicated, permission-scoped dashboard.

The platform solves three critical problems:

Problem	Solution
Students don't know what skills they lack	AI-powered Skill Gap Analyser with curated weekly learning paths
TPOs manage drives manually via spreadsheets	Shared-state Kanban board with automated status tracking
Recruiters struggle to find college talent	Bi-directional Drive Request system (Model A + Model B handshake)

Core Features by Role

Student Portal

Feature	Description
Profile Builder	Multi-step profile wizard: personal info, skills, CGPA, resume upload

Skill Gap Analyser	AI analysis against target domain/role using Gemini API
Learning Path Tracker	Milestone-based weekly study plans with progress tracking
Milestone Quizzes	AI-generated per-milestone quizzes; passing auto-completes milestone
Resume Upload & ATS	Upload to Cloudinary; ATS Reality Check scores resume vs job descriptions
Campus Drive Portal	View active college drives, eligibility check, one-click apply
Interview Prep	Access interview experiences and mock interview sessions
Job Opportunities	Smart job matching with skill-based percentage match scores
Profile Score	Dynamic placement readiness score based on profile completeness

College (TPO) Portal

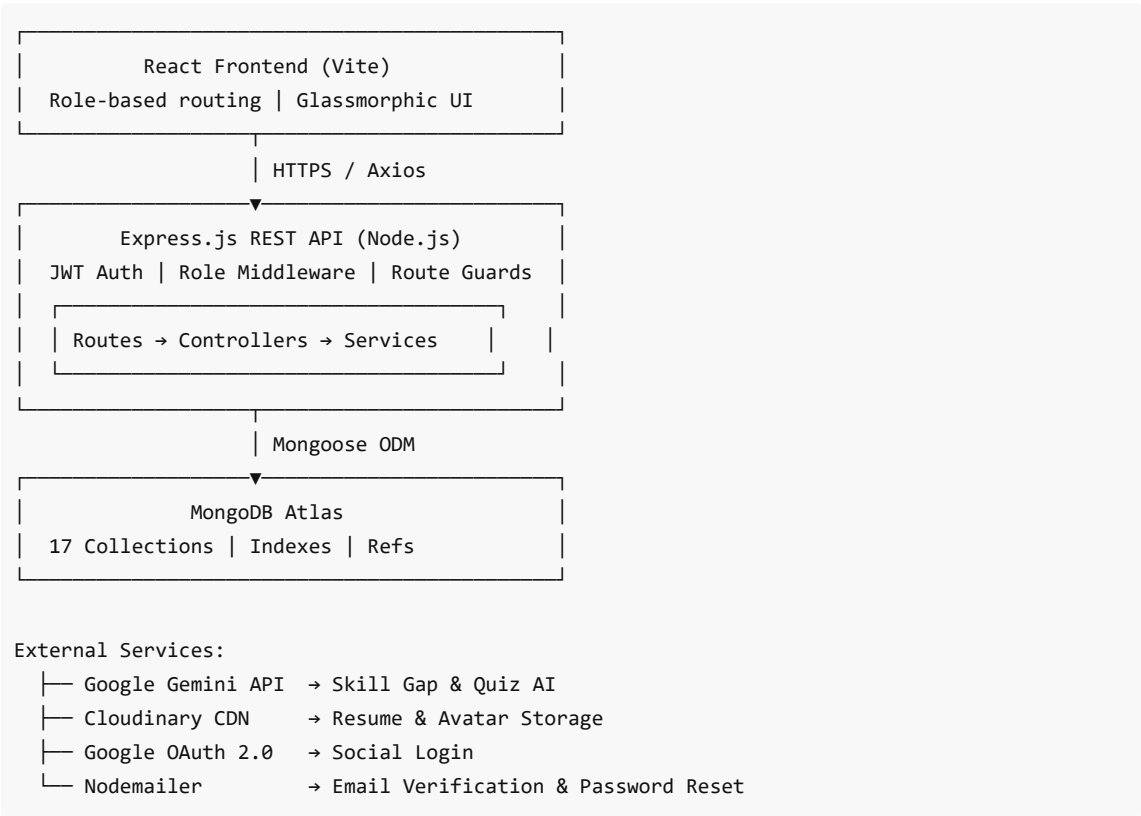
Feature	Description
College Dashboard	Macro analytics: placement rate, department breakdown, active drives
Campus Drives (Kanban)	Full 6-stage Kanban: Applied → Aptitude → Technical → HR → Offered → Rejected
Drive Creation	Create drives with eligibility (min CGPA), assign a specific recruiter
Incoming Drive Requests	Accept recruiter-initiated invites; auto-provisions a new drive on accept
Student Management	View and track all enrolled students and their placement readiness
Recruiter CRM	Track partner recruiter relationships, visit history, hire rates
Analytics	Placement success rates by department and academic year

Recruiter Portal

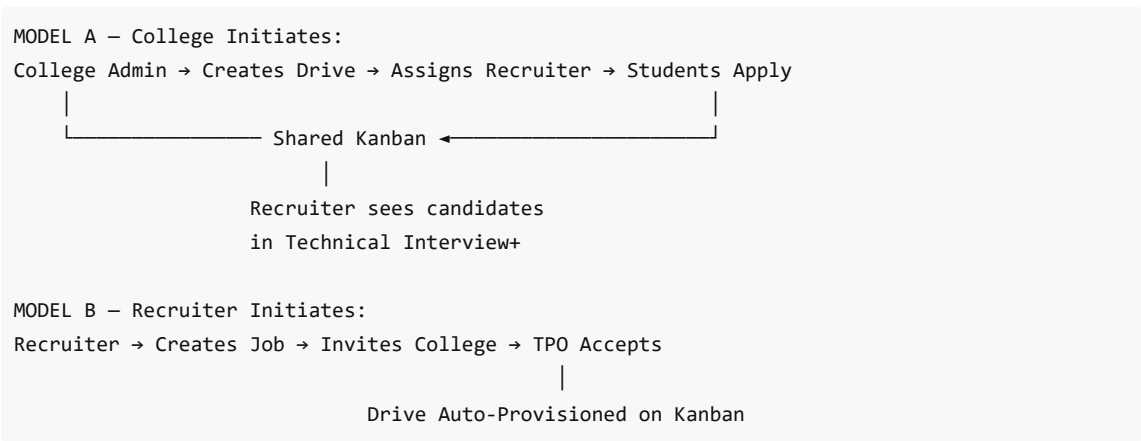
Feature	Description
Recruiter Dashboard	Overview of active drives, shortlisted candidates, open positions
Jobs & Postings	Create job postings with requirements; manage the full listing lifecycle
Invite Colleges (Model B)	Send campus drive invites to colleges directly from a job posting
Assigned Drives (Kanban)	Shared-state Kanban showing candidates pushed from Technical Interview onwards
Smart Shortlister	Filter students across all colleges by skills, CGPA, and department
College Explorer	Browse partner colleges, view placement stats, initiate drive requests
Saved Candidates	Bookmark promising candidates for later review
Domain Experts	Access a directory of verified industry mentors

System Architecture

SkillBridge uses a **3-tier MVC architecture**:



Campus Drive Data Flow (Hybrid Model)



Technology Stack

Frontend

Technology	Version	Purpose
React	18.x	UI Framework

Vite	5.x	Build tool & dev server
React Router DOM	v6	Client-side routing
Tailwind CSS	3.x	Utility-first styling
@hello-pangea/dnd	latest	Drag & Drop Kanban
Axios	latest	HTTP client
Lucide React	latest	Icon system

Backend

Technology	Version	Purpose
Node.js	18.x	Runtime
Express.js	4.x	Web framework
Mongoose	7.x	MongoDB ODM
JSON Web Token	latest	Stateless authentication
Bcrypt.js	latest	Password hashing
Passport.js	latest	OAuth 2.0 strategies
Multer + Cloudinary	latest	File upload pipeline
Nodemailer	latest	Transactional email
Google Generative AI	latest	Gemini API integration

Database & Cloud

Service	Purpose
MongoDB Atlas	Primary database (cloud)
Cloudinary	Resume & image CDN
Vercel / Railway	Deployment targets

Database Schema

Models Overview

Model	Collection	Key Fields
User	users	name, email, password, role, company, avatar
Student	students	user, college, skills[], cgpa, resumeUrl, targetRole
College	colleges	name, location, accreditation, totalStudents

Job	jobs	title, company, requirements[], salary, postedBy
CampusDrive	campusdrives	college, recruiter, jobId, title, eligibility, status
DriveApplication	driveapplications	student, drive, status (Kanban stage)
DriveRequest	driverequests	job, recruiter, college, status (pending/accepted/rejected)
SkillGapAnalysis	skillgapanalyses	student, targetDomain, gapSkills[], matchScore
SkillLearningPath	skilllearningpaths	student, skillName, milestones[], progressPercentage
DomainSkillRequirement	domainskillrequirements	domain, role, requiredSkills[]
MockInterview	mockinterviews	student, questions[], feedback
InterviewExperience	interviewexperiences	student, company, role, experience
Alumni	alumni	student, company, role, graduationYear
SavedCandidate	savedcandidates	recruiter, student, notes
JobMarketTrend	jobmarkettrends	domain, trendingSkills[], demandScore
Application	applications	student, job, status, appliedAt
StudentSkill	studentskills	student, skillName, proficiencyLevel, verified

API Reference

Authentication — /api/auth

Method	Endpoint	Auth	Description
POST	/register	✗	Register new user (any role)
POST	/login	✗	Login, returns JWT
POST	/forgot-password	✗	Send reset email
POST	/reset-password/:token	✗	Reset password via token
POST	/verify-email	✗	Verify email with OTP
GET	/google	✗	Google OAuth initiation
GET	/google/callback	✗	Google OAuth callback

Student Features — /api/student-features

Method	Endpoint	Auth	Description
GET	/profile	✓	Get student profile
PUT	/profile	✓	Update student profile

POST	/upload-resume	✓	Upload PDF resume to Cloudinary
GET	/resume-view	✓	Redirect to Cloudinary resume URL
POST	/upload-avatar	✓	Upload profile photo
GET	/drives	✓	Get active drives for student's college
POST	/drives/:id/apply	✓	Apply to a campus drive
GET	/job-opportunities	✓	Get skill-matched job listings
GET	/alumni	✓	Get alumni mentor directory
GET	/interview-experiences	✓	Browse interview experiences

Skill Gap Module — /api/skill-gap

Method	Endpoint	Auth	Description
POST	/analyze	✓ Student	Trigger AI skill gap analysis
GET	/latest	✓ Student	Get latest analysis result
GET	/history	✓ Student	Get analysis history (last 10)
GET	/domains	✗	List all available target domains & roles
GET	/learning-paths	✓ Student	Get all learning paths (deduplicated)
PATCH	/learning-paths/:id/progress	✓ Student	Update milestone progress
POST	/learning-paths/:id/reschedule	✓ Student	Reschedule overdue milestones
POST	/learning-paths/:id/milestones/:idx/quiz/generate	✓ Student	AI-generate milestone quiz
POST	/learning-paths/:id/milestones/:idx/quiz/submit	✓ Student	Submit quiz, auto-complete on pass

College Features — /api/college-features

Method	Endpoint	Auth	Description
GET	/dashboard	✓ College	Get dashboard stats & KPIs
GET	/students	✓ College	Get all enrolled students
GET	/analytics	✓ College	Placement analytics by department

GET	/recruiters	✓ College	Get all registered recruiters
GET	/drive-requests	✓ College	Get incoming recruiter drive invites
PUT	/drive-requests/:id/accept	✓ College	Accept invite → auto-create drive
GET	/crm	✓ College	Get recruiter CRM relationship data

Campus Drives — /api/campus-drives

Method	Endpoint	Auth	Description
GET	/	✓ College	Get all drives for the college
POST	/	✓ College	Create a new campus drive
GET	/:id/applications	✓ College	Get Kanban applications for a drive
PUT	/:id/applications/:appId/status	✓ College	Move application to new Kanban stage

Recruiter Features — /api/recruiter-features

Method	Endpoint	Auth	Description
GET	/candidates	✓ Recruiter	Smart shortlist with filters
GET	/saved-candidates	✓ Recruiter	Get bookmarked candidates
POST	/save-candidate	✓ Recruiter	Bookmark a student
GET	/colleges	✓ Recruiter	Browse college network
GET	/drives	✓ Recruiter	Get all assigned drives
GET	/drives/:id/applications	✓ Recruiter	Get Kanban apps for assigned drive
PUT	/drives/:id/applications/:appId/status	✓ Recruiter	Update candidate Kanban stage
POST	/drive-requests	✓ Recruiter	Send college drive invite for a job

Jobs — /api/jobs

Method	Endpoint	Auth	Description
GET	/	✗	List all job postings
POST	/	✓ Recruiter	Create new job posting
GET	/me	✓ Recruiter	Get jobs posted by current recruiter
GET	/opportunities	✓ Student	Get skill-matched opportunities

Other Routes

Route Base	Endpoint	Description
------------	----------	-------------

/api/ats	POST /check	ATS resume vs JD scoring
/api/interviews	GET /, POST /	Mock interview sessions
/api/analytics	GET /overview	Platform-wide analytics

Frontend Routing

All routes are managed by **React Router DOM v6** in `client/src/App.jsx`.

Public Routes

Path	Component	Description
/	LandingPage	Marketing homepage
/about	AboutUs	About the platform
/login	SignInPage	Unified login page
/signin	SignInPage	Alias for login
/signup	RoleSelectionPage	Choose account type
/signup/student	StudentSignup	Student registration
/signup/college	CollegeSignup	College registration
/signup/recruiter	RecruiterSignup	Recruiter registration
/signup/verify-email	EmailVerificationPage	OTP verification
/forgot-password	ForgotPasswordPage	Password reset request
/reset-password/:token	ResetPasswordPage	Set new password
/oauth-callback	OAuthCallbackPage	Google OAuth redirect handler

College (TPO) Routes — Role: college_admin

Path	Component	Description
/college	CollegeDashboard	Main TPO dashboard
/college/drives	CampusDrives	Kanban drive manager + incoming requests
/college/recruiters	RecruiterCRM	Partner recruiter management

Recruiter Routes — Role: recruiter

Path	Component	Description
/recruiter	RecruiterDashboard	Recruiter home dashboard
/recruiter/jobs	RecruiterJobs	Job postings + college invite system

/recruiter/drives	RecruiterDrives	Assigned drives Kanban view
/recruiter/smart-shortlist	SmartShortlist	Candidate filtering tool

Student Routes — Role: student

Path	Component	Description
/student	StudentDashboard	Full student intelligence hub
/student/drives	StudentDrives	Campus drive discovery & apply

Authentication & Security

JWT Flow

```
User Login → Server validates credentials → Issues signed JWT (7 days expiry)
↓
Client stores JWT in localStorage
↓
Every request: Header { "x-auth-token": "<JWT>" }
↓
Server middleware decodes → attaches req.user = { userId, role, collegeId }
↓
roleCheck(["college_admin"]) middleware enforces RBAC on protected routes
```

Security Measures

- **Password Hashing:** Bcrypt with salt rounds (12)
- **JWT Secret:** Stored in .env , never exposed
- **Role-Based Access Control (RBAC):** Every sensitive route uses roleCheck([roles]) middleware
- **Email Verification:** OTP-based before account activation
- **Cooldown Protection:** Skill gap analysis has a 5-minute cooldown to prevent API abuse
- **Drive Request Deduplication:** Server rejects duplicate invites for the same job+college pair
- **Google OAuth:** Passport.js strategy; JWT issued after callback (no sessions)

AI & Intelligence Layer

Gemini AI Integration

The platform uses **Google Gemini API** for:

Feature	Prompt Input	AI Output
Skill Gap Analysis	Student skills + target domain/role	Gap skills, recommendations, match score
Learning Path Generation	Gap skills, experience level	Weekly milestone schedule with resources

Milestone Quiz Generation	Skill name + milestone title	5 MCQs with answer keys
ATS Resume Scoring	Resume text + Job description	Score (0-100) + improvement tips

Smart Job Matching Algorithm

```
For each student:
  studentSkills = profile.skills[].skillName (normalized, lowercase)

  For each job:
    matchCount = job.requirements.filter(req → studentSkills.includes(req))
    matchScore = (matchCount / totalRequirements) * 100

  Filter: matchScore >= 20% (threshold to exclude irrelevant jobs)
  Sort: descending by matchScore
```

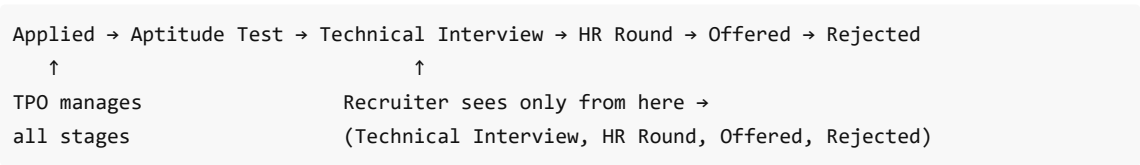
Skill Sync on Path Completion

When a student completes 100% of a learning path milestone:

- The skill is **automatically added** to their profile
- If the skill exists, its proficiency level is upgraded
- The skill is flagged as `verified: true` with source `"Skill Gap Analyser"`

Campus Drive Module

Kanban Stage Flow



Shared State Architecture

Both the **College TPO** and the **assigned Recruiter** operate on the same underlying `DriveApplication` documents. When either party drags a card:

1. The frontend fires a `PUT .../status` API call
2. The database record is updated
3. Both dashboards reflect the new state on next fetch

Drive Lifecycle — Model A (College-Initiated)

1. TPO creates drive (company, title, eligibility, minCGPA)
2. TPO optionally assigns a Recruiter from the dropdown
3. Students from the same college see the drive in their portal
4. Students who meet CGPA threshold can apply

5. TPO manages Kanban from Applied → Technical Interview
6. Recruiter takes over from Technical Interview → Offered/Rejected

Drive Lifecycle — Model B (Recruiter-Initiated)

1. Recruiter creates a Job Posting (title, requirements, salary)
2. Recruiter browses College Explorer → clicks "Invite to Drive"
3. Selects the job → sends DriveRequest (status: "pending")
4. TPO sees the invite in the "Incoming Drive Requests" panel
5. TPO clicks "Accept & Create Drive" → CampusDrive auto-provisioned
6. Both parties now share the Kanban board

Local Development Setup

Prerequisites

- Node.js v18+
- npm v9+
- MongoDB (Local or Atlas URI)
- Google Gemini API Key
- Cloudinary Account
- Google OAuth Credentials (optional for social login)

1. Clone & Install

```
# Clone the repository
git clone https://github.com/Guddu2222/skill_gap-Analyzer-and-placement_intelligence.git
cd "Skill_Gap Analyser and placement intelligence"

# Install all dependencies (root, server, client)
npm install
cd server && npm install
cd ../client && npm install
```

2. Configure Environment

Create `server/.env` :

```
PORT=5000
MONGODB_URI=mongodb+srv://<user>:<password>@cluster.mongodb.net/placement_platform
JWT_SECRET=your_super_secret_jwt_key_here
FRONTEND_URL=http://localhost:5173

# Google Gemini AI
GEMINI_API_KEY=your_gemini_api_key

# Cloudinary (File Uploads)
CLOUDINARY_CLOUD_NAME=your_cloud_name
CLOUDINARY_API_KEY=your_api_key
CLOUDINARY_API_SECRET=your_api_secret
```

```
# Google OAuth (Social Login)
GOOGLE_CLIENT_ID=your_google_client_id
GOOGLE_CLIENT_SECRET=your_google_client_secret
GOOGLE_CALLBACK_URL=http://localhost:5000/api/auth/google/callback

# Email (Password Reset & Verification)
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_USER=your_email@gmail.com
EMAIL_PASS=your_app_password
```

3. Run Development Server

```
# From the project root (runs both frontend + backend concurrently)
npm run dev
```

Service	URL
Frontend (Vite)	http://localhost:5173
Backend API	http://localhost:5000/api

Environment Variables

Variable	Required	Description
MONGODB_URI	✓	MongoDB connection string
JWT_SECRET	✓	Secret key for JWT signing
GEMINI_API_KEY	✓	Google Gemini AI API key
CLOUDINARY_CLOUD_NAME	✓	Cloudinary cloud name
CLOUDINARY_API_KEY	✓	Cloudinary API key
CLOUDINARY_API_SECRET	✓	Cloudinary API secret
FRONTEND_URL	✓	Frontend origin for CORS
PORT	✗	Server port (default: 5000)
GOOGLE_CLIENT_ID	✗	For Google OAuth
GOOGLE_CLIENT_SECRET	✗	For Google OAuth
EMAIL_USER	✗	For email verification
EMAIL_PASS	✗	App password for email

Deployment

Backend (Railway / Render)

1. Connect the GitHub repository
2. Set root directory to `server/`
3. Add all environment variables
4. Build command: `npm install`
5. Start command: `node index.js`

Frontend (Vercel)

1. Connect the GitHub repository
2. Set root directory to `client/`
3. Framework preset: **Vite**
4. Build command: `npm run build`
5. Output directory: `dist`
6. Add environment variable: `VITE_API_URL=https://your-backend.railway.app/api`

Project Structure

```
Skill_Gap Analyser and placement intelligence/
├── client/                                # React + Vite Frontend
│   ├── src/
│   │   ├── components/
│   │   │   ├── Sidebar.jsx              # Role-aware navigation
│   │   │   ├── HeroSection.jsx         # Landing hero
│   │   │   └── recruiter/              # Recruiter-specific widgets
│   │   │       ├── AdvancedSearch.jsx
│   │   │       ├── CollegeExplorer.jsx
│   │   │       ├── DomainExperts.jsx
│   │   │       └── SavedCandidates.jsx
│   │   └── student/
│   │       └── LearningPathTracker.jsx
│   └── pages/
│       ├── LandingPage.jsx
│       ├── SignInPage.jsx
│       ├── StudentDashboard.jsx         # Student intelligence hub
│       ├── StudentDrives.jsx          # Campus drive portal
│       ├── CollegeDashboard.jsx
│       ├── CampusDrives.jsx           # TPO Kanban + incoming requests
│       ├── RecruiterDashboard.jsx
│       ├── RecruiterJobs.jsx          # Job postings + college invite
│       ├── RecruiterDrives.jsx        # Assigned drives Kanban
│       ├── SmartShortlist.jsx
│       ├── RecruiterCRM.jsx
│       └── signup/                    # Multi-step registration
│           ├── RoleSelectionPage.jsx
│           ├── StudentSignup.jsx
│           ├── CollegeSignup.jsx
│           └── RecruiterSignup.jsx
```

```
|
└─ server/                                # Express.js Backend API
    │ index.js                            # Entry point + route bindings
    │ config/
    │ │ passport.js                      # Google OAuth strategy
    │ │ └─ cloudinary.js                  # Multer-Cloudinary pipeline
    │ │
    │ └─ middleware/
    │ │ └─ auth.js                       # JWT verify + roleCheck()
    │ │
    │ └─ models/                         # 17 Mongoose schemas
    │ │ └─ User.js
    │ │ └─ Student.js
    │ │ └─ College.js
    │ │ └─ Job.js
    │ │ └─ CampusDrive.js
    │ │ └─ DriveApplication.js
    │ │ └─ DriveRequest.js
    │ │ └─ SkillGapAnalysis.js
    │ │ └─ SkillLearningPath.js
    │ │ └─ ...
    │ │
    │ └─ routes/                         # 10 route files
    │ │ └─ auth.js
    │ │ └─ jobs.js
    │ │ └─ campus-drives.js
    │ │ └─ college-features.js
    │ │ └─ recruiter-features.js
    │ │ └─ student-features.js
    │ │ └─ skillGap.js
    │ │ └─ interviews.js
    │ │ └─ ats.js
    │ │ └─ analytics.js
    │ │
    │ └─ services/
    │ │ └─ skillGapAnalysis.service.js    # Gemini AI orchestration
```

Design System

The frontend uses the "**Aetheris Abyssal**" design system — a premium dark glassmorphism aesthetic:

- **Primary Background:** #0c0c1d (deep space)
- **Surface Layers:** #121223 → #1d1d33 → #24233b (elevation via tonal shift)
- **Accent:** Indigo #a3a6ff + Violet #ac8aff
- **Typography:** Inter (Google Fonts), weight range 400–900
- **UI Style:** Glassmorphism (backdrop-blur , rgba fills), no hard borders
- **Animations:** 0.3s ease-out transitions, pulsing glows on CTAs

Built with ❤️ — SkillBridge | Intelligent Placement for the Next Generation